

2024



Building a Generative AI Contact Center Solution for DoorDash Using Amazon Bedrock, Amazon Connect, and Anthropic’s Claude

Learn how DoorDash built a generative AI self-service contact center solution using Amazon Bedrock, Amazon Connect, and Anthropic’s Claude.

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50x increase

in testing capacity

50% reduction

in response latency

2.5 seconds or less

response latency with Anthropic's Claude 3 Haiku

100Ks of calls

per day fielded by generative AI solution



Hi, I can connect you with an AWS representative or answer questions you have on AWS.



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



50% reduction

in generative AI application development time using Amazon Bedrock



Overview

[DoorDash](#) receives hundreds of thousands of requests for assistance through its contact center from Consumers, Merchants, and Dashers—independent contractors who deliver through the platform—each day.

To streamline support, DoorDash wanted to harness the power of [generative artificial intelligence \(AI\)](#) to enhance its self-service offerings and elevate the user experience. DoorDash collaborated alongside Amazon Web Services (AWS) through the [AWS Generative AI Innovation Center](#) (GenAIIC) program, which pairs companies with AWS experts to implement generative AI solutions, to build a fully voice-operated self-service generative AI contact center solution that was ready for live testing in only 2 months.



Opportunity | Building a Generative AI Contact Center Solution to Elevate User Experience for Millions of Dashers Globally

DoorDash started in 2013 with the mission to grow and empower local economies—helping local businesses succeed, conveniently connecting local consumers to the best in their neighborhood, and providing Dashers with flexible ways to earn. By the end of 2023, its user base had grown to more than 37 million active consumers who use DoorDash each month and over 2 million monthly active Dashers.

To meet the needs of Consumers, Merchants, and Dashers, DoorDash uses [Amazon Connect](#), an AI-powered contact center from AWS. Through Amazon Connect, DoorDash handles a collective volume of hundreds of thousands of calls per day, including from Dashers who need assistance with routine topics ranging from app troubleshooting to sign-ups and payment options.

Upon contacting support, users are guided through a self-service interactive voice response (IVR) experience, powered by Amazon Connect and [Amazon Lex](#), which has reduced DoorDash's agent transfers by 49 percent and driven a 12 percent increase in first contact resolution leading to a better experience for DoorDash users, and \$3M in YoY operational cost savings. However, with most calls still being redirected to live agents, opportunity to further enhance its self-service offering. "We wanted to empower Dashers to c



most common questions and issues as quickly and efficiently as possible, saving them time, effort, and increasing their trust in DoorDash's self-service capabilities,” says Chaitanya Hari, Contact Center Product Lead at DoorDash.

Dashers generally prefer calling into support rather than chatting while they’re on the road driving to or from Merchant or Consumer locations. They rely on phone support to provide fast and reliable assistance, which makes response latency in any self-service solution doubly important. DoorDash needed to make sure that Dashers can spend as little time on the phone as possible, so low response latency became a key factor for its phone solution.

To streamline Dasher support, DoorDash’s contact center team wanted to use generative AI for self-service within Amazon Connect. The team sought a solution it could roll out quickly and at scale while maintaining a high standard for issue resolution and customer satisfaction. The team chose [Amazon Bedrock](#), a fully managed service that offers a choice of high-performance foundation models from leading AI companies through a single API, to serve as the base of its solution.

Using AWS, we’ve built a solution that gives Dashers reliable access to the information they need, when they need it.”

Chaitanya Hari

Contact Center Product Lead, DoorDash

Solution | Supercharging DoorDash’s IVR Experience Using Generative AI Services

The AWS GenAIIC invested in and worked with DoorDash to design, build, and deploy a custom generative AI solution to enhance its existing voice AI assistant. Over 8 weeks, teams from DoorDash and the GenAIIC iterated over design and implementation and developed a reference architecture that was suitable for production A/B testing. The Amazon Bedrock–based solution reduced generative AI application development time by 50 percent.

To build its solution, DoorDash chose to use [Anthropic's Claude models in Amazon Bedrock](#). Using Amazon Bedrock, DoorDash quickly got access to all the models it needed. Claude was instrumental to the project because it has the capability to mitigate hallucinations, prompt injection events, and detect abusive language. With the release of Claude 3 Haiku, DoorDash achieved the accuracy and speed it needed for its voice application, achieving a response latency of 2.5 seconds or less. DoorDash does not provide any personally identifiable information to be accessed via these generative AI solutions. In addition, [Amazon Bedrock enables data security](#) through encryption and ensures data from DoorDash or its customers are only used in the DoorDash application.

“Employing AI-generated responses for phone support presented unique challenges that demanded innovative strategies to enhance response times and answer quality, providing Dashers with best-in-class support.”



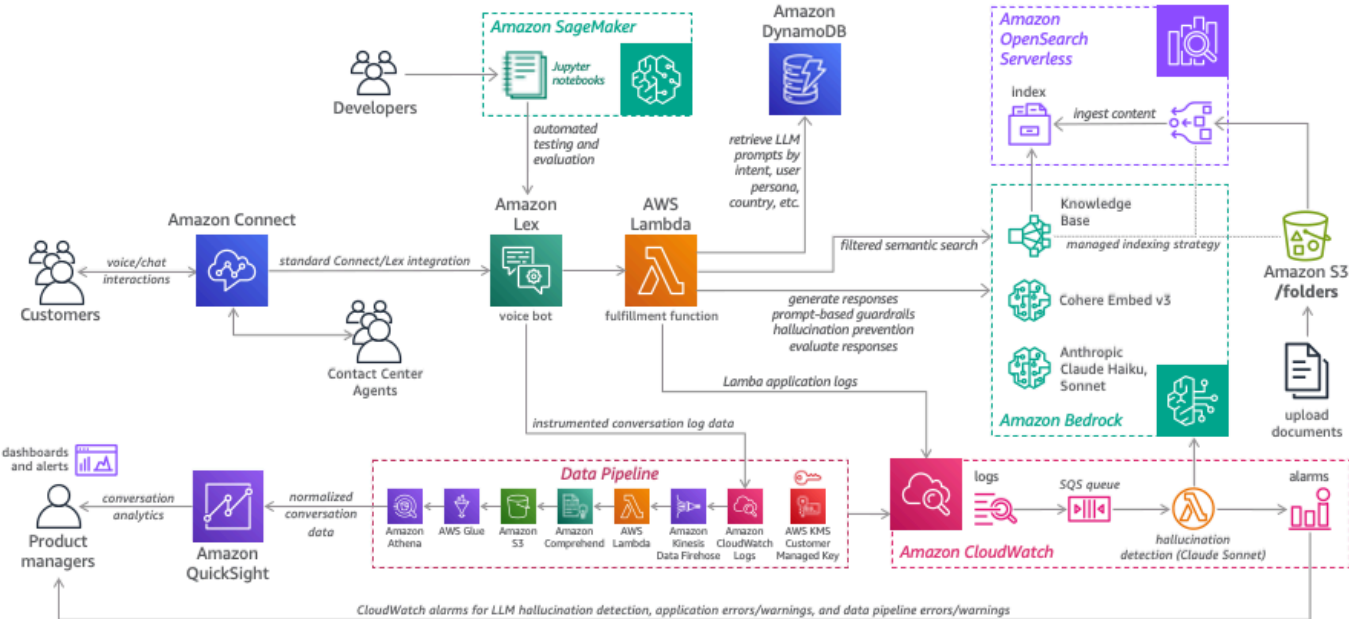
Bedrock proved to be a perfect fit for our requirements, allowing us to concentrate on refining the solution's finer details," says Vraj Shah, Lead Project Engineer at DoorDash.

Access to a deep knowledge base was another important factor for the self-service solution. The more in-depth and diverse the knowledge base, the more effective the solution. DoorDash added data from its publicly available help center to use [retrieval-augmented generation \(RAG\)](#), a technique that fetches data from company sources and enriches the prompt to provide more relevant and accurate responses for Dashers.

To index existing content, it used [Knowledge Bases for Amazon Bedrock](#), a fully managed capability that helps organizations implement the entire RAG workflow from ingestion to retrieval and prompt augmentation without having to build custom integrations to data sources and manage data flows. Knowledge Bases for Amazon Bedrock took care of the backend work for DoorDash.

Figure 1. Contact center RAG solution architecture with conversation analytics

Full end-to-end contact center RAG solution architecture



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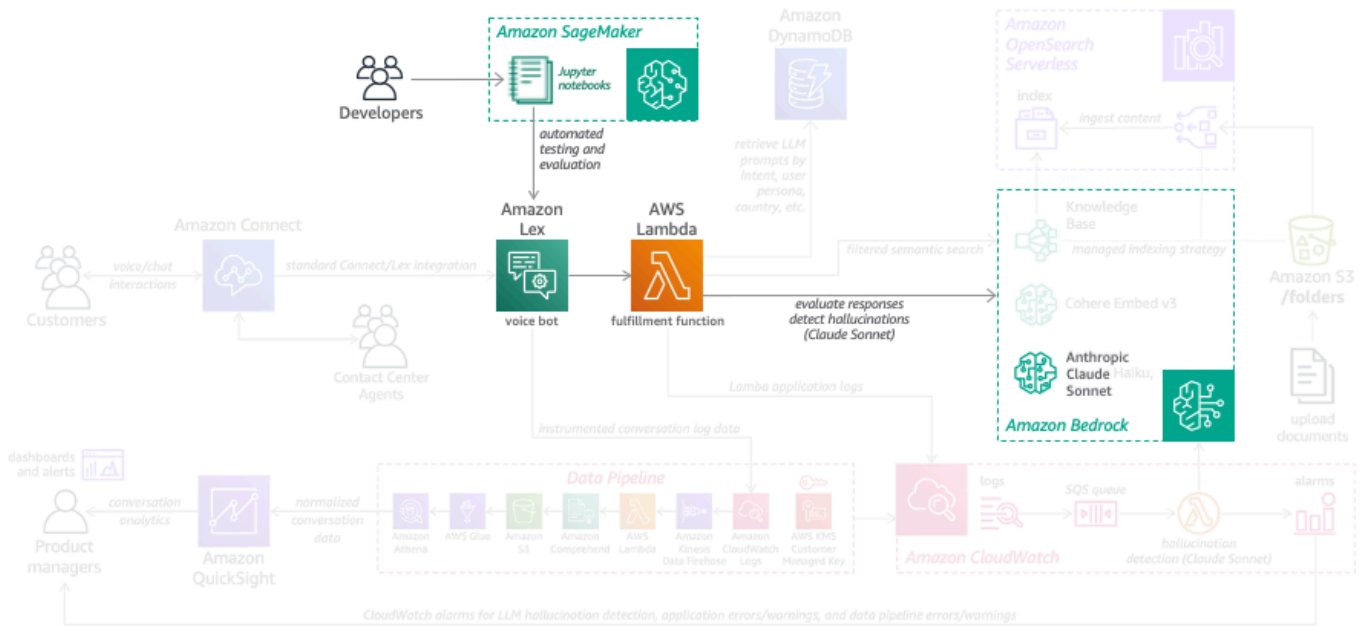
DoorDash also increased the capacity to test its solution with help from the GenAIIC. Previously, the team had to pull contact center agents off help queues to complete a small number of manual test cases per hour. Using [Amazon SageMaker](#), which developers use to build, train, and deploy machine learning models, DoorDash built a test and evaluation framework that quickly drew insights from A/B testing and evaluated key success metrics at scale. This framework helps DoorDash to complete thousands of automated tests per hour—a 50x increase in capacity—and semantically evaluates responses against ground-truth data.

By automatically handling common Dasher inquiries, DoorDash has improved self-service workflows and increased issue resolution speeds, which helps to accelerate delivery time and enhance Dasher productivity and satisfaction. Addressing routine questions with generative AI has also freed up DoorDash’s live agents to solve higher-complexity issues that benefit from human troubleshooting.

Figure 2. Automated testing



Automated testing



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Outcome | Launching DoorDash's Solution and Expanding to New Use Cases

After completing a successful test of the generative AI solution in early 2024, DoorDash has since completed the rollout of the new self-service options for all Dashers. The solution is currently fielding hundreds of thousands of Dasher support calls each day and has driven large and material reductions in call volumes for Dasher-related support inquiries. The solution has also reduced the number of escalations to live agents by thousands per day, and reduced the number of live agent tasks required to solve support inquiries.

Given the early success of the rollout, the team is currently working on adding more functionality to the solution, expanding the breadth of the available knowledge bases for the solution to draw from and incorporating DoorDash's event-driven logistics workflow service to not only provide question and answer assistance but also to take actions on behalf of the user.

“Using AWS and Anthropic’s Claude, we’ve built a solution that gives Dashers reliable and simple-to-understand access to the information they need, when they need it,” says Hari. “This has cascading positive impacts on our users and the platform as a whole, and we look forward to expanding to new use cases in the future.”

About DoorDash

DoorDash is a local commerce platform dedicated to helping Merchants thrive in the convenience economy, giving consumers access to more of their communities, and providing work that empowers.



Chaitanya Hari

Voice/Contact Center Product Lead, DoorDash

In his role, Chaitanya spearheads the strategic and technological advancement of DoorDash's voice systems, across applications in sales, support, and more. He strives to consistently push the boundaries of contact center technology for DoorDash's benefit, by leveraging the latest advancements in AI to enhance operational efficiencies, improving DoorDash's overall unit economics, and ultimately creating a stellar customer experience.



Vraj Shah

Connect Developer, DoorDash

AWS Services Used

Vraj is a developer on the Voice team for DoorDash. He has a strong background in AWS technologies with a focus on Amazon Connect and uses that to advance the contact center technology for DoorDash. He has designed and implemented solutions that have been the key to increase operational excellence, enhance agent productivity and improve customer satisfaction. These have helped provide a personalized and seamless experience to the customers, resulting in greater satisfaction for the customers and better cost-

Amazon Bedrock

Amazon Bedrock is a fully managed service that offers a choice of high-performing foundation models (FMs) from leading AI companies and a broad set of capabilities you need to build generative AI applications with security, privacy, and responsible AI.

[Learn more »](#)



Amazon Connect

Transform your customer experience (CX) at scale with Amazon Connect, an AI-powered contact center from AWS.

[Learn more »](#)

Amazon Lex

Amazon Lex is a fully managed artificial intelligence (AI) service with advanced natural language models to design, build, test, and deploy conversational interfaces in applications.

[Learn more »](#)

Amazon SageMaker

Amazon SageMaker is a fully managed service that brings together a broad set of tools to enable high-performance, low-cost machine learning (ML) for any use case.

[Learn more »](#)

AWS Generative AI Innovation Center

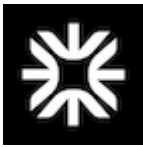
A program that pairs you with AWS science and strategy experts with deep experience in AI/ML and generative AI techniques to imagine new applications of generative AI to address your needs, identify new use cases based on business value, and integrate Generative AI into your existing applications and workflows.

[Learn more »](#)

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